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JEE Main Physics Practice Sheet — DPP-037

Topic: Kinetic Energy, Work-Energy Theorem & Practical Applications

Time Allowed: 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

- Q1.** If the linear momentum of a body increases by 50%, the percentage increase in its kinetic energy is:
- (A) 50%
(B) 125%
(C) 100%
(D) 225%
- Q2.** A bullet of mass m penetrates into a fixed wooden block and loses half of its velocity after traversing a distance $d = 3$ cm. Assuming a constant resistive force, the additional distance the bullet will penetrate before coming to complete rest is:
- (A) 1 cm
(B) 3 cm
(C) 2 cm
(D) 1.5 cm
- Q3.** A block of mass $m = 2$ kg is dropped from a height $h = 40$ cm onto a vertical spring of spring constant $k = 1960$ N/m. The maximum compression x of the spring is ($g = 9.8$ m/s²):
- (A) 0.05 m
(B) 0.20 m
(C) 0.10 m
(D) 0.15 m
- Q4.** The Work-Energy Theorem state that the net work done on a particle equals the change in its kinetic energy ($W_{\text{net}} = \Delta K$). This theorem is valid in:
- (A) Inertial frames only
(B) Non-inertial frames only if work done by pseudo forces is included
(C) Both inertial and non-inertial frames (with pseudo force work accounted for)
(D) Conservative force fields only
- Q5.** A particle of mass $m = 0.5$ kg moves along a straight line with velocity $v(x) = 5x^{3/2}$ m/s. The work done by the net force during its displacement from $x = 0$ to $x = 2$ m is:
- (A) 200 J
(B) 50 J
(C) 100 J
(D) 400 J
- Q6.** A block of mass $m = 4$ kg sliding on a frictionless horizontal floor at $v = 10$ m/s enters a rough patch of length $L = 5$ m where the kinetic friction coefficient varies as $\mu_k(x) = 0.1x$. The speed of the block as it exits the rough patch is ($g = 10$ m/s²):
- (A) 5.0 m/s
(B) 8.66 m/s
(C) 7.07 m/s
(D) 9.22 m/s
- Q7.** A body of mass $m = 2$ kg is projected vertically upward with speed $u = 30$ m/s. If air resistance does -100 J of work during the upward ascent, the maximum height reached by the body is ($g = 10$ m/s²):
- (A) 40 m
(B) 45 m
(C) 35 m
(D) 50 m
- Q8.** A car of mass M moving at speed v is brought to rest over a minimum stopping distance s by maximum braking force. If the initial speed of the car is tripled ($v' = 3v$) under identical road conditions, the new minimum stopping distance will be:
- (A) $3s$
(B) $6s$
(C) $9s$
(D) $\sqrt{3}s$
- Q9.** A variable force $F = 6t$ N acts on a body of mass $m = 2$ kg initially at rest on a frictionless surface. The work done by the force in the first $t = 2$ seconds is:
- (A) 18 J
(B) 36 J
(C) 72 J
(D) 9 J
- Q10.** Two masses $m_1 = 4$ kg and $m_2 = 9$ kg have equal kinetic energies. The ratio of their linear momenta p_1/p_2 is:
- (A) 2 : 3
(B) 4 : 9
(C) 3 : 2
(D) 16 : 81
- Q11.** A particle moves along the x -axis from $x = 0$ to $x = 4$ m under the influence of force $F(x) = (10 - 2x)$ N. If its kinetic energy at $x = 0$ is 16 J, its kinetic energy at $x = 4$ m is:
- (A) 24 J
(B) 32 J
(C) 40 J
(D) 48 J

Q12. A chain of mass m and length l is released from rest on a smooth table with $1/3$ rd of its length hanging over the edge. The speed of the chain as its last link slips off the table is:

- (A) $\sqrt{\frac{8gl}{9}}$
- (B) $\sqrt{\frac{2gl}{3}}$
- (C) $\sqrt{\frac{4gl}{9}}$
- (D) $\sqrt{\frac{gl}{3}}$

Q13. A pendulum bob of mass m is suspended by a string of length L . The minimum horizontal speed u that must be imparted to the bob at the lowest point so that the string becomes slack only after reaching the horizontal level ($\theta > 90^\circ$) is:

- (A) $\sqrt{2gL}$
- (B) $\sqrt{5gL}$
- (C) $\sqrt{3gL}$
- (D) $\sqrt{gL}$

Q14. A block of mass $m = 1$ kg is attached to a horizontal spring of stiffness $k = 100$ N/m. The block is compressed by $x_0 = 0.2$ m and released from rest on a rough table ($\mu_k = 0.1$). The speed of the block as it passes through the natural length position is ($g = 10$ m/s²):

- (A) 1.89 m/s
- (B) 2.00 m/s
- (C) 1.50 m/s
- (D) 1.73 m/s

Q15. An engine pumps water continuously through a hose with cross-sectional area A . If water emerges with speed v and density ρ , the rate at which kinetic energy is imparted to the water is:

- (A) $\frac{1}{2}\rho Av^3$
- (B) ρAv^3
- (C) $\frac{1}{2}\rho Av^2$
- (D) $\frac{1}{4}\rho Av^3$

Q16. A body of mass m is moving in a circular path of radius r with a constant tangential acceleration a_t . If the kinetic energy of the body is K after n complete revolutions from rest, the value of a_t is:

- (A) $\frac{K}{4\pi nmr}$
- (B) $\frac{K}{2\pi nmr}$

- (C) $\frac{K}{mr}$
- (D) $\frac{4\pi nK}{mr}$

Q17. A particle of mass m is moving in a force field where potential energy is $U(r) = \frac{a}{r^2} - \frac{b}{r}$ ($a, b > 0$). The kinetic energy of the particle in a stable circular orbit of radius r_0 is:

- (A) $\frac{b^2}{8a}$
- (B) $\frac{b^2}{4a}$
- (C) $\frac{b^2}{2a}$
- (D) $\frac{b}{2r_0}$

Q18. A bullet of mass m travelling horizontally with velocity v strikes a sandbag of mass M suspended by a rope of length L and gets embedded in it. The loss of kinetic energy during the collision is:

- (A) $\frac{1}{2} \left(\frac{Mm}{M+m} \right) v^2$
- (B) $\frac{1}{2}mv^2$
- (C) $\frac{1}{2} \left(\frac{m^2}{M+m} \right) v^2$
- (D) $\frac{1}{2} \left(\frac{M^2}{M+m} \right) v^2$

Q19. A uniform rope of length L and mass M held vertically is dropped onto a table from rest with its lower end just touching the tabletop. The kinetic energy of the rope when a length y has landed on the table is:

- (A) $\frac{1}{2}Mg \left(\frac{L-y}{L} \right) y$
- (B) $Mgy \left(\frac{L-y}{L} \right)$
- (C) $\frac{1}{2}Mgy$
- (D) $\frac{1}{2}Mg(L-y)$

Q20. A particle of mass m begins to move along the x -axis under a net force such that its power delivered varies with time as $P(t) = \alpha t$ ($\alpha = \text{constant}$). The displacement s of the particle from rest as a function of time t is proportional to:

- (A) t^2
- (B) $t^{3/2}$
- (C) $t^{1/2}$
- (D) t^3

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-037)

Kinetic Energy, Work-Energy Theorem & Practical Applications

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	B	6	B	11	C	16	A
2	A	7	A	12	A	17	A
3	C	8	C	13	A	18	A
4	C	9	B	14	A	19	B
5	C	10	A	15	A	20	A

Detailed Step-by-Step Solutions

Sol 1. Option (B)

Kinetic energy in terms of linear momentum:

$$K = \frac{p^2}{2m}$$

When momentum increases by 50%: $p' = 1.5p$.

$$K' = \frac{(1.5p)^2}{2m} = 2.25 \left(\frac{p^2}{2m} \right) = 2.25K$$

Percentage increase:

$$\frac{K' - K}{K} \times 100\% = (2.25 - 1) \times 100\% = 125\%$$

Sol 2. Option (A)

Let initial speed be u and constant retarding force be F_R .
Phase 1: Speed drops from u to $u/2$ over distance $d = 3$ cm:

$$W_1 = -F_R d = \frac{1}{2} m \left(\frac{u}{2} \right)^2 - \frac{1}{2} m u^2 = -\frac{3}{8} m u^2 \Rightarrow F_R(3) = \frac{3}{8} m u^2 \Rightarrow F_R = \frac{1}{8} m u^2$$

Phase 2: Speed drops from $u/2$ to 0 over additional distance x :

$$W_2 = -F_R x = 0 - \frac{1}{2} m \left(\frac{u}{2} \right)^2 = -\frac{1}{8} m u^2 \Rightarrow F_R x = \frac{1}{8} m u^2 \Rightarrow x = 1 \text{ cm}$$

Sol 3. Option (C)

Let x be the maximum compression of the spring. Total vertical fall = $h + x$.

By Work-Energy Theorem from release to maximum compression ($\Delta K = 0$):

$$W_{\text{gravity}} + W_{\text{spring}} = 0 \Rightarrow mg(h + x) - \frac{1}{2} k x^2 = 0$$

$$2(9.8)(0.4 + x) - \frac{1}{2}(1960)x^2 = 0 \Rightarrow 19.6(0.4 + x) - 980x^2 = 0$$

$$7.84 + 19.6x - 980x^2 = 0 \Rightarrow 50x^2 - x - 0.4 = 0$$

$$x = \frac{1 \pm \sqrt{1 - 4(50)(-0.4)}}{100} = \frac{1 \pm \sqrt{1 + 80}}{100} = \frac{1 + 9}{100} = 0.10 \text{ m}$$

Sol 4. Option (C)

The Work-Energy Theorem ($W_{\text{net}} = \Delta K$) is universally valid in all inertial frames, as well as non-inertial reference frames provided the work done by pseudo forces is included in W_{net} .

Sol 5. Option (C)

At $x = 0$, $v_i = 0$. At $x = 2$ m, $v_f = 5(2)^{3/2}$ m/s $\Rightarrow v_f^2 =$

$$25(2^3) = 200 \text{ m}^2/\text{s}^2.$$

By Work-Energy Theorem:

$$W = \Delta K = \frac{1}{2} m v_f^2 - 0 = \frac{1}{2} (0.5)(200) = 50 \text{ J}$$

Sol 6. Option (B)

Work done by variable friction over the rough patch:

$$W_f = - \int_0^5 \mu_k(x) mg dx = - \int_0^5 (0.1x)(4)(10) dx = -4 \int_0^5 x dx = -50 \text{ J}$$

By Work-Energy Theorem:

$$W_f = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 \Rightarrow -50 = \frac{1}{2} (4) v_f^2 - \frac{1}{2} (4)(10^2)$$

$$-50 = 2v_f^2 - 200 \Rightarrow 2v_f^2 = 150 \Rightarrow v_f = \sqrt{75} \approx 8.66 \text{ m/s}$$

Sol 7. Option (A)

By Work-Energy Theorem from projection to apex ($v_f = 0$):

$$W_{\text{gravity}} + W_{\text{air}} = 0 - \frac{1}{2} m u^2$$

$$-mgh - 100 = -\frac{1}{2} (2)(30^2) = -900 \Rightarrow -20h = -800 \Rightarrow h = 40 \text{ m}$$

Sol 8. Option (C)

$$\text{Frictional braking work: } -f_k s = 0 - \frac{1}{2} M v^2 \Rightarrow s = \frac{M v^2}{2 f_k} \propto v^2.$$

When initial speed is tripled ($v' = 3v$): $s' = 3^2 s = 9s$.

Sol 9. Option (B)

$$\text{Acceleration } a(t) = \frac{F}{m} = \frac{6t}{2} = 3t.$$

$$\text{Velocity at } t = 2 \text{ s: } v(2) = \int_0^2 3t dt = \left[\frac{3t^2}{2} \right]_0^2 = 6 \text{ m/s.}$$

By Work-Energy Theorem:

$$W = \Delta K = \frac{1}{2} m v^2 - 0 = \frac{1}{2} (2)(6^2) = 36 \text{ J}$$

Sol 10. Option (A)

Momentum in terms of kinetic energy: $p = \sqrt{2mK}$.

Since $K_1 = K_2 = K$:

$$\frac{p_1}{p_2} = \frac{\sqrt{2m_1 K}}{\sqrt{2m_2 K}} = \sqrt{\frac{m_1}{m_2}} = \sqrt{\frac{4}{9}} = \frac{2}{3}$$

Sol 11. Option (C)

Work done by force:

$$W = \int_0^4 (10 - 2x) dx = [10x - x^2]_0^4 = 40 - 16 = 24 \text{ J}$$

By Work-Energy Theorem:

$$K_f = K_i + W = 16 + 24 = 40 \text{ J}$$

Sol 12. Option (A)

Initial mass on table = $\frac{2}{3}m$, hanging mass = $\frac{1}{3}m$ with initial CM at $y_i = \frac{l/3}{2} = \frac{l}{6}$.

Initial PE: $U_i = -\left(\frac{m}{3}\right)g\left(\frac{l}{6}\right) = -\frac{mgl}{18}$.

Final PE (entire chain hanging just as it slips off, CM at $l/2$): $U_f = -mg\left(\frac{l}{2}\right) = -\frac{mgl}{2}$.

By Conservation of Energy / Work-Energy:

$$\Delta K = -\Delta U = U_i - U_f = -\frac{mgl}{18} - \left(-\frac{mgl}{2}\right) = \frac{mgl}{2} - \frac{mgl}{18} = \frac{8mgl}{18} = \frac{4mgl}{9}$$

$$\frac{1}{2}mv^2 = \frac{4}{9}mgl \Rightarrow v = \sqrt{\frac{8gl}{9}}$$

Sol 13. Option (A)

To reach the horizontal position ($\theta = 90^\circ$), velocity at horizontal position must be at least zero ($v \geq 0$).

Work done by gravity to reach horizontal: $W_g = -mgL$.

By Work-Energy Theorem:

$$-mgL = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 \Rightarrow u^2 = v^2 + 2gL \geq 2gL \Rightarrow u_{\min} = \sqrt{2gL}$$

Sol 14. Option (A)

From release ($x = 0.2 \text{ m}$, $v_i = 0$) to natural length ($x = 0$):

Work done by spring: $W_s = \frac{1}{2}kx_0^2 = \frac{1}{2}(100)(0.2)^2 = 2.0 \text{ J}$.

Work done by friction: $W_f = -\mu_k mgx_0 = -0.1(1)(10)(0.2) = -0.2 \text{ J}$.

By Work-Energy Theorem:

$$W_s + W_f = \frac{1}{2}mv^2 \Rightarrow 2.0 - 0.2 = \frac{1}{2}(1)v^2 \Rightarrow 1.8 = 0.5v^2 \Rightarrow v = \sqrt{3.6} = 1.8 \text{ m/s}$$

Sol 15. Option (A)

Mass ejected per second: $\frac{dm}{dt} = \rho Av$.

Kinetic energy imparted per unit time (Power):

$$\frac{dK}{dt} = \frac{1}{2} \left(\frac{dm}{dt} \right) v^2 = \frac{1}{2}(\rho Av)v^2 = \frac{1}{2}\rho Av^3$$

Sol 16. Option (A)

Total distance travelled in n revolutions: $s = n(2\pi r) = 2\pi nr$.

Tangential force: $F_t = ma_t$.

Work done: $W = F_t \cdot s = (ma_t)(2\pi nr) = 2\pi nmr a_t$.

By Work-Energy Theorem: $W = K \Rightarrow a_t = \frac{K}{2\pi nmr}$ or $\frac{K}{2\pi nmr}$.

Sol 17. Option (A)

Force: $F(r) = -\frac{dU}{dr} = -\left(-\frac{2a}{r^3} + \frac{b}{r^2}\right) = \frac{2a}{r^3} - \frac{b}{r^2}$.

For circular orbit: $\frac{mv^2}{r} = \frac{b}{r^2} - \frac{2a}{r^3} \Rightarrow mv^2 = \frac{b}{r} - \frac{2a}{r^2}$.

Equilibrium radius r_0 occurs where $F(r_0) = 0 \Rightarrow r_0 = \frac{2a}{b}$.

Kinetic energy: $K = \frac{1}{2}mv^2 = \frac{1}{2} \left(\frac{b}{r_0} - \frac{2a}{r_0^2} \right) = \frac{1}{2} \left(\frac{b^2}{2a} - \frac{9}{4a^2} \right) = \frac{1}{2} \left(\frac{b^2}{4a} \right) = \frac{b^2}{8a}$.

Sol 18. Option (A)

Initial KE: $K_i = \frac{1}{2}mv^2$.

Final common velocity: $V = \frac{mv}{M+m}$.

Final KE: $K_f = \frac{1}{2}(M+m)V^2 = \frac{1}{2} \frac{m^2 v^2}{M+m}$.

Loss in KE:

$$\Delta K = K_i - K_f = \frac{1}{2}mv^2 \left(1 - \frac{m}{M+m} \right) = \frac{1}{2} \left(\frac{Mm}{M+m} \right) v^2$$

Sol 19. Option (B)

The remaining moving part of the rope has mass $M' = M \left(\frac{L-y}{L} \right)$.

Since it falls freely under gravity through distance y , its speed is $v = \sqrt{2gy}$.

Kinetic energy:

$$K = \frac{1}{2}M'v^2 = \frac{1}{2} \left(M \frac{L-y}{L} \right) (2gy) = Mgy \left(\frac{L-y}{L} \right)$$

Sol 20. Option (A)

$P = \frac{dK}{dt} = \alpha t \Rightarrow K(t) = \int \alpha t dt = \frac{1}{2}\alpha t^2$.

Since $K = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{\alpha}{m}}t$.

Displacement $s = \int v dt = \sqrt{\frac{\alpha}{m}} \left(\frac{t^2}{2} \right) \propto t^2$.